MILITARY SPECIFICATION

SWIM FINS, RUBBER

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. - This specification covers the requirements for swim fins made of gum rubber for wear by military personnel for swimming purposes and for general utility.

1.2 Classification. - The swim fins shall be of one type in the following sizes (see 6.2):

Schedule of sizes

Swin Fin (sizes)	For Shoe Sizes
Medium-large	8 up to 9-1/2
Large	9-1/2 up to 11
Extra-large	11 up to 12-1/2
Super extra-large	12-1/2 up to 14

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal
UU-P-553 - Paper, Wrapping, Tissue
PPP-B-636- Box, Fiberboard

STANDARDS

Federal
FED. Test Method STD.NO. 601 - Rubber: Sampling and Texting

Military
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-177 - Rubber Products, Terms for Visible Defects of.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 8465

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2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing Materials

ASTM D412-61T	-	Tension Testing of Vulcanized Rubber.
ASTM D471-59T	-	Change in Properties of Elastomeric Vulcanizates Resulting from Immersion in Liquids.
ASTM D624-54	-	Tear Resistance of Vulcanized Rubber.
ASTM D1149-60T	-	Accelerated Ozone Cracking of Vulcanized Rubber.
ASTM D1630-61	-	Abrasion Resistance of Rubber Soles and Heels.

(Information regarding copies of ASTM methods of testing may be obtained from the American Society for Testing Materials, 1916 Race Street, Philadelphia 3, Pa.)

3. REQUIREMENTS

3.1 Sample. - Samples are furnished as a guide only. Variations from this specification may appear in the sample, in which case this specification shall govern (see 6.2 and 6.3).

3.2 Preproduction sample. - When specified (see 6.2), before production is commenced, a sample of the finished swim fins shall be submitted or made ready for the contracting officer or his authorized representative for inspection as specified in 4.2.1. The approval of the preproduction sample authorizes the commencement of production but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured by the supplier in the same facilities to be used for the manufacture of the production items.

3.3 Materials

3.3.1 Rubber compounds. - The polymer used in the manufacture of the swim fins shall be compounded from natural (gum) or natural and synthetic rubber for the flipper and natural rubber for the strap. The rubber compounds shall be made by such processes as to meet the physical requirements listed in table I when tested as specified in 4.4.

3.3.2 Color and finish. - The color of the finished swim fins shall be natural, unless otherwise specified (see 6.2). A light application of finish containing only a sufficient amount of pigment to assist in obtaining a uniform color shall be applied, and the surface brought to a smooth finish. The finished swim fins shall be nonmarking and shall produce only a slight, erasable mark on white paper, when tested as specified in 4.4.2, and shall not leave a permanent mark when scuffed.

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Table I. - Physical requirements of swim fins

	Flipper	Strap
Rubber compound	Natural or natural and synthetic	Natural
Initial (standard conditions):		
Tensile strength, psi	2800 min.	2900 min.
Modulus		
300 percent, psi	1000 min.	180 min.
500 percent, psi	1800 min.	450 min.
Elongation at break, percent	600 min.	750 min.
Permanent set, percent (at 75 percent E_B)	80 max.	4 max.
Tear strength (Die C), lbs per inch	290 min.	150 min.
Abrasive index (N.B.S.), revolutions per 0.10 inch loss	350 min.	---
Specific gravity	0.98 (±.02)	0.96 (±.02)
Hardness Shore A:		
Before aging, points	65 (± 5)	40 (± 5)
After aging, points	75 (± 5)	50 (± 5)
Aged conditions:		
Volume swell oil resistance, 7 days room temperature		
Fuel A, volume change, percent	125 max.	260 max.
Fuel B, volume change, percent	180 max.	350 max.
Static bent-loop ozone resistance, 50 pphm (small cracks appearance), hours	30	15

3.4 Marking. - Marking shall be distinctly molded on each swim fin with 3/8 inch high block letters. The inscriptions shall be legible and centered along top and bottom sides approximately 1/2 inch from edges of swim fins at locations shown in figure 1.

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3.4 Marking (cont'd)

Top side inscription

SWIM FINS
(SIZE) SUPER EXTRA-LARGE (Example)

Bottom side inscription

U. S. GOVT. PROPERTY
(CONTRACT NO.) DSA-1-123-66A(Example)

3.4.1 Instruction label. - Each pair of swim fins shall contain a printed paper label approximately 3-1/2 inches long and 2-1/2 inches wide, bearing the following:

INSTRUCTIONS FOR MAINTENANCE AND CARE
After using in salt water, SWIM FINS should be rinsed and hung up to dry. DO NOT EXPOSE OR STORE in direct rays of sun nor in a heated locker or storage area. Heavy objects should not be placed on top of the SWIM FINS for long periods of time to prevent any permanent set. CAUTION: SWIM FINS are usually worn one size larger than the shoe size of the swimmer to allow for flipper slipper, boot or other covering that is worn to give warmth to the foot and for protection against heel strap abrasions.

3.5 Design. - The molded swim fins shall be of commercial design conforming to the type and sizes as specified (see 1.2, 3.6, and figure 1).

3.6 Molds. - The molds used for the manufacture of the swim fins shall be furnished by the supplier. The molds used shall include allowance for the overlap of the natural rubber strap during molding with the flipper of natural-synthetic rubber compound into a 1-piece material swim fin as specified in 3.3.1.

3.7 Figure. - The figure is furnished for information purposes only. To the extent of any inconsistencies between the written specification and the figure, the written specification shall govern.

3.8 Sizes and measurements. - Sizes and measurements of the finished swim fins shall be as shown in table II. All measurements and tolerances are expressed in inches, and shall conform to the dimensions at the locations designated (see figure 1).

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Table II. - Measurements of swim fins

SIZE	L E N G T H S					
	Overall		Flipper		Inside Foot	
	Fin (A)1/	Strap (B)1/	Top (C)1/	Bottom (D)1/	Top (E)1/	Bottom (F)1/
Medium-large (8 up to 9-1/2)	21-1/4	3-1/2	17-3/4	16-5/8	6-13/16	5-11/16
Large (9-1/2 up to 11)	21-3/4	3-5/8	18-1/8	16-7/8	7-3/16	6
Extra-large (11 up to 12-1/2)	22-1/4	3-3/4	18-1/2	17-1/8	7-9/16	6-5/16
Super extra-large (12-1/2 up to 14)	22-3/4	3-7/8	18-7/8	17-3/8	7-15/16	6-5/8
Tolerance	± 1/2	± 1/4	± 1/2	± 1/2	± 1/8	± 1/8

Table II. - Measurements of swim fins (cont'd)

SIZE	W I D T H S			H E I G H T S			
	Toe	Instep		Outside		Inside	
	(G)1/	Top (H)1/	Bottom (I)1/	(J)1/	(K)1/	Arch (L)1/	Toe (M)1/
Medium-large (8 up to 9-1/2)	8-13/16	3-1/4	3-9/16	2-11/16	7/8	2-13/16	15/16
Large (9-1/2 up to 11)	9	3-5/16	3-11/16	2-13/16	15/16	2-15/16	1
Extra-large (11 up to 12-1/2)	9-3/16	3-7/16	3-13/16	2-7/8	15/16	3-1/16	1
Super extra-large (12-1/2 up to 14)	9-3/8	3-1/2	3-7/8	2-15/16	1	3-3/16	1-1/16
Tolerance	± 1/4	± 1/8	± 1/8	± 1/8	± 1/16	± 1/8	± 1/16

1/ See figure 1.

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A. Taken at top, from point of flipper to end of strap
B. Taken at top, from flipper instep to end of strap
C. Taken at top, from point of flipper to edge of instep
D. Taken at bottom, from point of flipper to edge of instep
E. Taken at top center along inside edge of instep to inside end of instep at top
F. Taken at bottom center along inside edge of instep to inside end of instep at bottom
G. Taken at top, across flipper, edge to edge, at widest points
H. Taken at top, along instep edge
I. Taken at bottom, along instep edge
J. Taken at side, from bottom of instep on a straight line to top surface of side rib
K. Taken at side, from bottom to top on a straight line to top of toe rib
L. Taken on inside height from ankle opening to inside sole level
M. Taken on inside toe height, 1/2 inch from end of toe pocket

3.9 Workmanship. - The swim fins shall be clean and smooth, and free from defects which affect appearance or which may impair serviceability. The finished swim fins shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preacceptance inspection

4.2.1 Preproduction sample inspection. - When required, the preproduction sample submitted in accordance with 3.2, shall be inspected for compliance with all requirements of this specification.

4.2.2 Inspection of components and materials. - Quality assurance for components and materials described, shall be in accordance with this specification or subsidiary specifications referenced herein to the extent applicable. In addition the rubber compounds shall meet the requirements specified in table I and 3.3.1 when tested as specified in table III and 4.4.2

4.3 Inspection for acceptance. - Sampling and inspection shall be performed in accordance with MIL-STD-105, except as otherwise specified.

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4.3.1 Inspection of the end item

4.3.1.1 Examination of the end item. - The finished swim fins shall be examined for visual and dimensional defects. The defects found shall be classified in accordance with 4.3.1.2 and 4.3.1.3. The inspection levels and acceptable quality levels (AQL's) shall be as specified in 4.3.1.4.

4.3.1.2 Visual examination. - The swim fins shall be examined for applicable defects listed in MIL-STD-177. Defects found shall be counted regardless of the proximity of each other except where two or more defects represent a single local condition in which case only the more serious defect shall be counted. The swim fins shall be examined for defects in pairing, design, material, workmanship, finish and marking. The sample unit shall be one completely fabricated swim fin but the sample units shall be selected as pairs. For pairing examination the pair of swim fins shall be examined together. The lot size shall be expressed in units of finished swim fins.

Examine	Defect	Classification	
		Major	Minor
Pairing	Not properly mated, i.e., not of same size, wide variation in color or appearance	X	
Cleanliness	Any spot, stain or streak clearly noticeable		X
	Flash or fin protrusions not clearly trimmed		X
Color and finish	Not specified color	X	
	Color not uniform; streaky		X
	Surface finish not uniformly smooth		X
	Flipper and strap not smoothly joined		X
Design	Other than specified	X	
	Misshaped	X	
Marking			
Swim fins	Missing, incomplete, incorrect or illegible	X	
	Not applied in the specified manner or misplaced		X
Instruction label	Missing, incomplete, incorrect or illegible		X

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Examine	Defect	Classification Major	Minor
Material			
General	Not fabricated of the material specified	X	
	Any cut, tear, hole, repair or other type of weakening damage	X	
	Non-weakening damages		X
Rubber	Backflash--distortion along mold line	X	
	Blister--extending through body	X	
	Charge, stop or heat mark	X	
	Crack--extending through body	X	
	Deformed; any part badly twisted	X	
	Any porous area of air check	X	
	Any area of sponginess	X	
	Any separations between rubber parts, between surfaces, or along rises on mold line	X	
	Frosting or flow marks		X
	Large particles of imbedded foreign matter		
	Affecting serviceability	X	
	Numerous small imbedded particles		X
	Overcured i.e., brittle or stiffness	X	
	Undercured i.e., evidence of tacky area or logginess	X	

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4.3.1.3 Examination for dimensional defects. - Any dimension that is not within the tolerance shown in table II shall be scored as a dimensional defect. The sample unit shall be one completely fabricated swim fin. The lot size shall be expressed in terms of the sample unit.

4.3.1.4 Inspection levels and acceptable quality levels (AQL's). - The acceptable quality levels expressed in defects per hundred units and the inspection levels shall be as follows:

	Inspection level	AQL
For defects applicable to 4.3.1.2	II	2.5 Major, 6.5 total (Major and Minor Combined)
For defects applicable to 4.3.1.3	S-2	4.0

4.4 Testing of the end item

4.4.1 Testing of end item rubber components. - The methods of testing rubber materials specified in ASTM Standards, FED.STD. 601 and table III wherever applicable shall be followed for end item testing. When the data in the "Number of determinations per sample unit" and the "Results reported as" columns are not specified in table III they shall be as required by the referenced test method. The physical and chemical values specified in section 3 represent the average of the number of test specimens selected for a sample unit for test purposes as specified in the applicable test methods. The sample unit for testing for each of the rubber compounds shall be a 6 by 6 inch slab. The number of sample units required for testing shall be as follows: The lot shall be unacceptable if one or more units fail to meet any requirement specified.

Lot size	Sample size
800 or less	2
801 to 22,000	3
22,001 and over	5

4.4.2 Nonmarking test. - The edge of the finished swim fin shall be drawn firmly across a sheet of white paper. If marking occurs, note whether it is easily erased by rubbing with the fingers (see 3.3.1).

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Table III. - Instructions for testing

Characteristic	Requirement paragraph	Test method	Applicable to: Sample unit	Applicable to: Lot avg.	No. deter. per sample unit	Results reported as
Identification of rubber compound						
Flipper (Natural and Synthetic)	3.3.1 & Table I	Certificate of compliance 1/	X	-	1	Pass or fail
Strap (natural)	3.3.1 & Table I	Certificate of compliance 1/	X	-	1	Pass or fail
Nonmarking	3.3.1	(see 4.4.2)	X	-	3	Pass or fail
Tensile strength	Table I	ASTM-D412-61T	-	X	3	To nearest pound
Modulus--						
300 percent	Table I	ASTM-D412-61T	-	X	3	To nearest pound
500 percent	Table I	ASTM-D412-61T	-	X	3	To nearest pound
Elongation	Table I	ASTM-D412-61T	-	X	3	To nearest percent
Permanent set	Table I	ASTM-D412-61T	-	X	3	To nearest percent
Tear strength	Table I	ASTM-D624-54	-	X	3	To nearest pound
Abrasive index	Table I	ASTM-1630-59T	-	X	3	To nearest unit
Specific gravity	Table I	FED.STD.601 - Method 14011		X	3	To nearest unit
Hardness (Shore A) (Before and after aging)	Table I	FED.STD.601 Methods 3021 & 7221 2/	-	X	5	Whole scale reading

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Table III. - Instructions-for testing (cont'd)

Characteristic	Requirement paragraph	Test method	Applicable to: Sample unit	Applicable to: Lot avg.	No. deter. per sample unit	Results reported as
After Aging:						
Volume swell (oil resistance)						
Fuel A	Table I	ASTM-D471-59T	X	-	3	To nearest percent
Fuel B	Table I	ASTM-D471-59T	X	-	3	To nearest percent
Ozone resistance	Table I	ASTM-D1149-60T				Pass or fail

1/ The contractor shall furnish a certificate of compliance stating that the rubber compounds conform to the requirements specified.

2/ Aging shall be conducted at 212° (± 4°)F., or at 100° (± 2.2°)C., for 24 hours.

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4.4.3 Examination for preparation of delivery. - An examination shall be made to determine that the packaging, packing and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per hundred units.

Examine	Defect
Markings (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence or method of application.
Materials	Any component missing or not as specified. Any component damaged, affecting serviceability.
Workmanship	Inadequate application of components, such as: incomplete closure of container flaps, loose strapping, inadequate stapling. Bulging or distortion of containers.
Contents	Number of packages per shipping container is more or less than required. Number of items per package is more or less than required.1/ Sizes shown on swim fins not the same as specified on packages or shipping containers.1/

1/ For this defect one package shall be examined from each shipping container in the sample.

5. PREPARATION FOR DELIVERY

5.1 Packaging. - Packaging shall be level A or C (see 6.2).

5.1.1 Level A. - Each swim fin shall be completely wrapped with machine glazed (lint free) tissue paper conforming to type I, grade A, class 1 or 2 of UU-P-553. Each pair of wrapped swim fins, of one size only, with each swim fin reversed end to end, shall be packaged in a two-piece telescope box conforming to style FTC, type CF, class-domestic, variety SW, grade 175 of PPP-B-636. The inside dimensions of each box shall be 23 inches in length, 9-1/2 inches in width, and 4-3/4 inches in depth.

5.1.2 Level C. - Swim fins shall be packaged in accordance with the industry's practice.

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5.2 Packing. - Packing shall be level A, B, or C (see 6.2).

5.2.1 Level A. - Six pairs of swim fins, of one size only, packaged as specified in 5.1.1, shall be packed in a fiberboard shipping container conforming to style RSC, grade V3c or V3s of PPP-P-636. The six packages shall be packed flat, one in length, two in width, and three in depth within the shipping container. The inside dimensions of the shipping container shall be approximately 24 inches in length, 20-1/2 inches in width, and 15-1/2 inches in depth. The approximate dimensions are furnished as a guide only. Exact dimensions to be furnished by the contractor. Each shipping container shall be closed, waterproofed and reinforced with flat steel strapping or tape banding in accordance with the appendix of the container specification.

5.2.2 Level B. - Six pairs of swim fins, of one size only, packaged as specified in 5.1.1, shall be packed in a fiberboard shipping container conforming to style RSC, type CF or SF, class-domestic, variety SW, grade 275 of PPP-B-636. The six packages shall be packed flat, one in length, two in width, and three in depth within the shipping container. The inside dimensions of the shipping container shall be approximately 24 inches in length, 20-1/2 inches in width, and 15-1/2 inches in depth. The approximate dimensions are furnished as a guide only. Exact dimensions to be furnished by the contractor. Each shipping container shall be closed in accordance with method II of the appendix of the container specification.

5.2.3 Level C. - Swim fins shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Shipping containers shall be in accordance with the rules or regulations of carriers applicable to the mode of transportation.

5.3 Marking. - In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129. Also, the following information shall be added on the exterior container:

STORE IN COOL, DRY PLACE

6. NOTES

6.1 Intended use. - The swim fins covered by this specification are intended for use by authorized military personnel as a swimming accessory either with or without self-contained underwater breathing apparatus. Maximum thrust is obtained with the flutter kick at normal rate of 30 kicks per minute per foot which will enable the swimmer to make headway, with equipment, against a current of 1-1/2 to 2 knots.

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6.2 Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Sizes required (see 1.2).
c. Availability of samples (see 3.1).
d. Whether preproduction sample is required (see 3.2).
e. Whether color other than natural is required (see 3.3.2).
f. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Sample. - For access to samples of swim fins address the procuring office issuing the invitation for bids.

Custodians:

Army - GL
Navy - SA
Air Force - 67

Preparing Activity:

Navy - SA
(NAVSUPRANDFAC)

Project No. 8465-0176

Interested Activities:

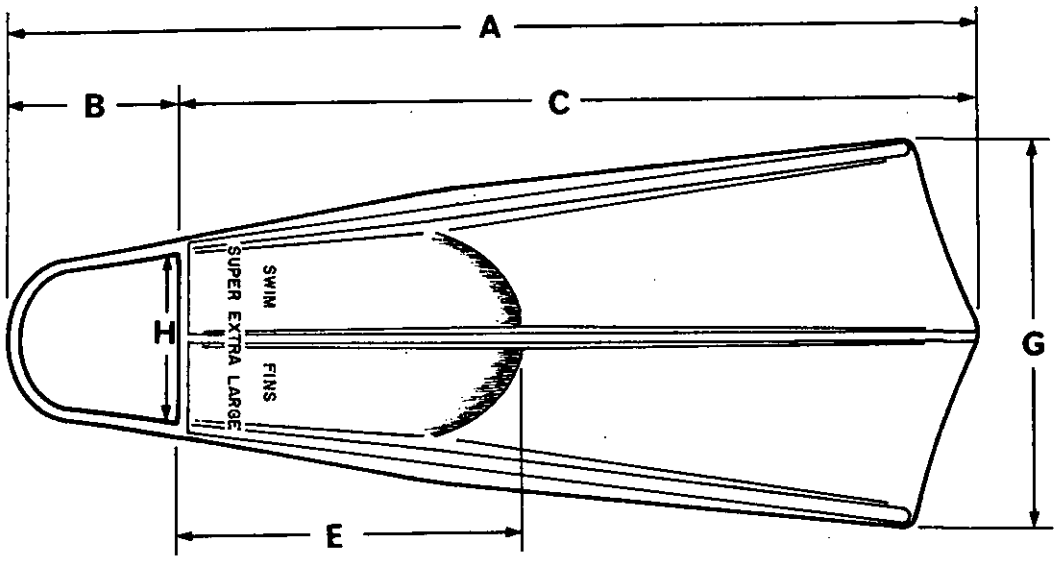
Review:
Army - GL, MD
Navy - SA, BUSHIPS
Air Force - 67

User:

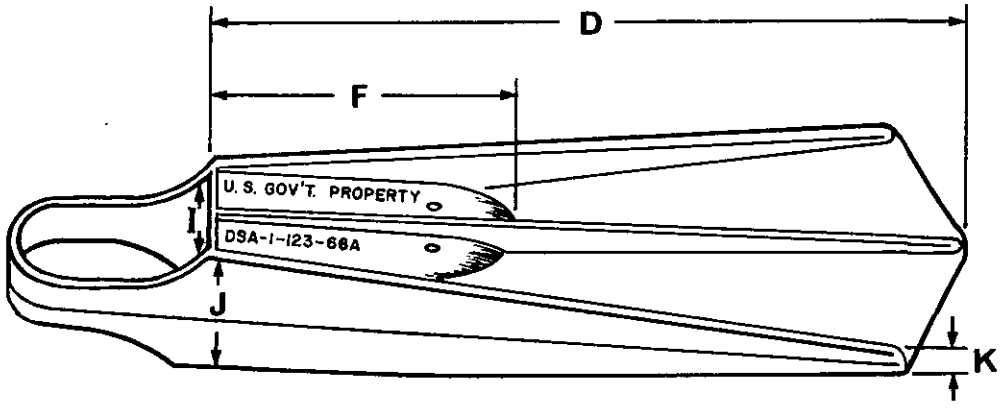
Navy - MC, CG

Reviewer and User information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current Federal Supply Classification Listing of DOD Standardization Documents.

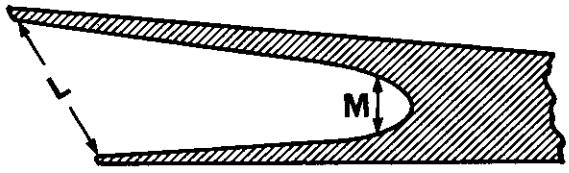
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TOP VIEW



BOTTOM VIEW



CROSS SECTION VIEW

FIGURE 1 - SWIM FINS, RUBBER

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

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